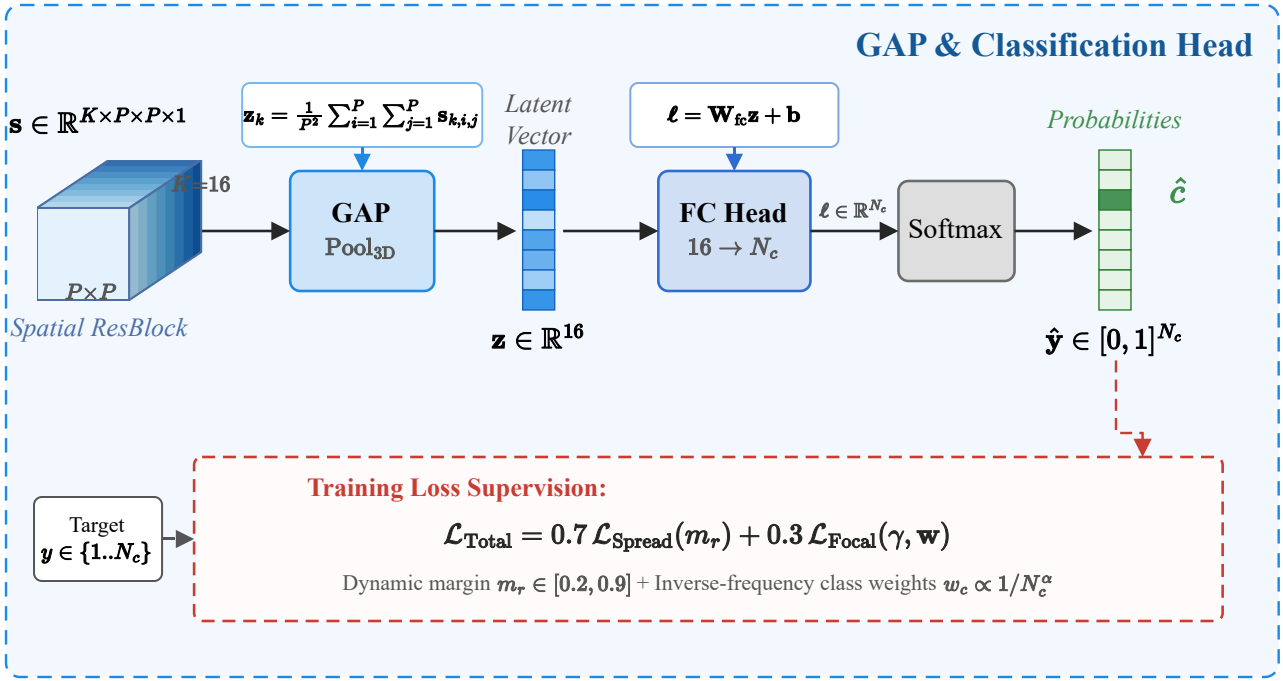

GAP & Classification Head
$\mathbf{s} \in \mathbb{R}^{K\times P\times P\times 1}$
K=16
$P\times P$
Spatial ResBlock
$\mathbf{z}_k = \frac{1}{P^2}\sum_{i=1}^{P}\sum_{j=1}^{P}\mathbf{s}_{k,i,j}$
GAP
Pool_{3D}
Latent Vector
$\mathbf{z} \in \mathbb{R}^{16}$
$\boldsymbol{\ell} = \mathbf{W}_{\text{fc}}\mathbf{z} + \mathbf{b}$
FC Head
$16 \to N_c$
$\boldsymbol{\ell} \in \mathbb{R}^{N_c}$
Softmax
Probabilities
$\hat{\boldsymbol{c}}$
$\hat{\mathbf{y}} \in [0,1]^{N_c}$
Target
$y \in \{1..N_c\}$
Training Loss Supervision:
$\mathcal{L}_{\text{Total}} = 0.7\,\mathcal{L}_{\text{Spread}}(m_r) + 0.3\,\mathcal{L}_{\text{Focal}}(\gamma, \mathbf{w})$
Dynamic margin $m_r \in [0.2, 0.9]$ + Inverse-frequency class weights $w_c \propto 1/N_c^{\alpha}$